

University of Al-Kafeel College of Dentistry



LEC-1

General Principles of Pharmacology

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Learning Objectives

By the end of this lecture, you should be able to

- Define Pharmacology and explain its branches.
- Understand basic pharmacokinetic and pharmacodynamic principles.
- Recognize commonly used drug classes in dental practice.
- Apply pharmacological knowledge to safe and effective dental treatment.



What is Pharmacology?



Pharmacology is the branch of biomedical science that studies the **interactions** between drugs and living organisms.

It explores how drugs affect biological systems and how the body responds to them.

It includes:

- **Pharmacodynamics** – what the drug does to the body (mechanism of action, effects)
- **Pharmacokinetics** – what the body does to the drug (absorption, distribution, metabolism, elimination)

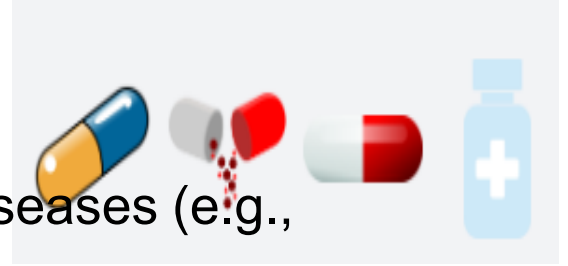
Why Pharmacology Matters in Dentistry



- 1. Pain Management** Dentists rely on pharmacological agents like **analgesics** (e.g., ibuprofen, acetaminophen) and **local anesthetics** (e.g., lidocaine) to control pain during and after procedures.
- 2. Infection Control** Understanding **antibiotics** (e.g., amoxicillin, metronidazole) is essential for treating or preventing oral infections, especially in immunocompromised or post-surgical patients.
- 3. Anesthesia and Sedation** Safe and effective use of **local anesthetics**, **sedatives**, and **general anesthetics** requires knowledge of drug interactions, metabolism, and patient-specific risks.

Why Pharmacology Matters in Dentistry

Cont



4. Management of Systemic Conditions Many dental patients have chronic diseases (e.g., diabetes, hypertension) and take medications that affect oral health or interact with dental drugs. Pharmacology helps dentists adjust treatment plans accordingly.

5. Emergency Alertness Dentists must be prepared to manage emergencies like anaphylaxis, hypoglycemia, or cardiac arrest, using drugs such as epinephrine, glucose, or nitroglycerin.

6. Drug Interactions and Side Effects A solid pharmacological foundation helps dentists avoid harmful interactions and recognize oral manifestations of systemic drug effects (e.g., gingival hyperplasia from phenytoin)

Branches of Pharmacology

Pharmacology is a broad discipline that explores how drugs interact with biological systems. It includes several specialized branches:



- **Pharmacodynamics** focuses on what drugs do to the body, including mechanisms of action, receptor interactions, and dose-response relationships.
- **Pharmacokinetics** examines what the body does to drugs—how they are absorbed, distributed, metabolized, and eliminated.
- **Clinical Pharmacology** applies pharmacological principles to patient care, emphasizing therapeutic efficacy, safety, and individualized treatment.
- **Toxicology** studies the harmful effects of chemicals and drugs, including overdose, poisoning, and organ-specific toxicity.



Question : All Drugs given to the body are considered safe ?

Drug

A **drug** is any chemical substance that, when administered to a living organism, **modifies one or more of its physiological or biochemical functions**.

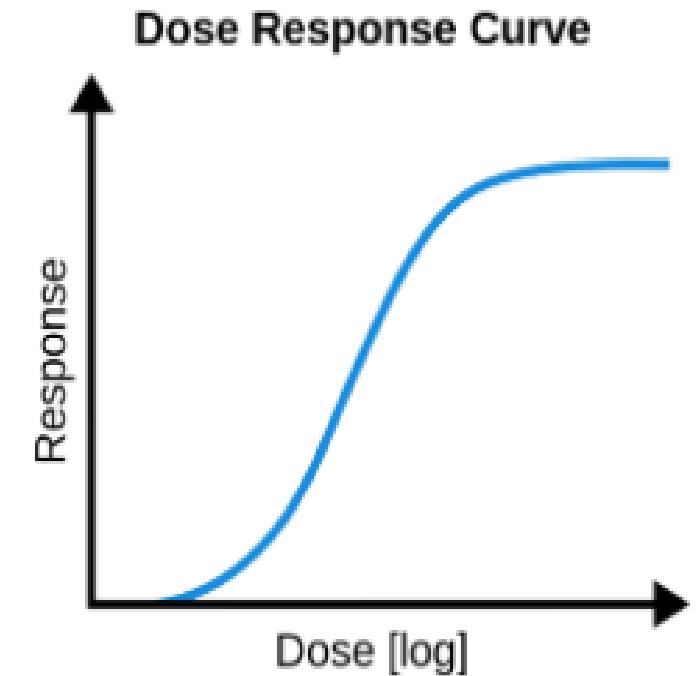
Drugs may be used for:

- **Diagnosis** (e.g., contrast agents)
- **Prevention** (e.g., vaccines)
- **Treatment or cure** of diseases (e.g., antibiotics, antihypertensives)
- **Relief of symptoms** (e.g., analgesics, antipyretics)



Characteristics of Drugs

- **Specificity**: Acts on particular cells, tissues, or receptors.
- **Dose-dependent effects**: Therapeutic vs. toxic outcomes.
- **Pharmacological activity**: Can be beneficial or harmful.
- **Can be natural, synthetic, or semi-synthetic.**



Source of the drugs

(1) Synthetic drugs:

Aspirin , Paracetamol, antidiabetics, antihistaminic

(2) Natural Drugs :

(a) Plant: morphine, codeine, quinine.

(b) Animal: gonadotropins, heparin, enzymes.

(c) Microorganisms: penicillin, streptomycin.

(d) Minerals: iron, calcium, Magnesium.



Source and nature of drugs contd.

(3) Semi synthetic drugs:

Ampicillin , Amoxicillin, Cloxacillin



(4) Biosynthetic drugs:

Produced by biological systems or engineered organisms

- Insulin
- Growth hormone,
- Hepatitis vaccine



Drug Names



Individual drugs may have several different names, but the two that are most commonly used are the generic name and the trade or brand name.

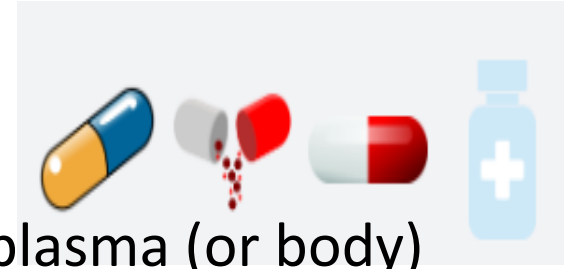
- A. **The Generic name** (e.g., amoxicillin) is related to the chemical or official name and is independent of the manufacturer. The generic name often indicates the drug group (e.g., drugs with generic names ending in “cillin” are penicillins).
- B. **The Trade or brand name** is designated and patented by the manufacturer. For example, Amoxicillin is manufactured by several pharmaceutical companies, some of which assign a specific trade name (e.g., Amoxil, Trimox) .

General Pharmacological Terms



- **Pharmacokinetics**: is the study of what the body does to the drug, i.e. absorption, distribution, metabolism and secretion (ADME) of the drug.
- **Pharmacodynamics**: is the study of what the drug does to the body, i.e. biological effects (pharmacological and toxic) and mechanism of action of drug.
- **METABOLISM (Biotransformation)**: metabolic processes convert lipid soluble drugs into water soluble products, which are easily eliminated either in the urine or through the bile.
- **Therapeutic Index (window) TI**: the difference between the minimum effective dose of the drug and the minimum toxic dose of the drug, between which the therapeutic benefit occur

Half-life ($t_{1/2}$)



Half-life ($t_{1/2}$) is the time required for the concentration of a drug in the plasma (or body) to **decrease** by **50%** due to metabolism and elimination. It reflects how long a drug stays active in the body and helps determine:

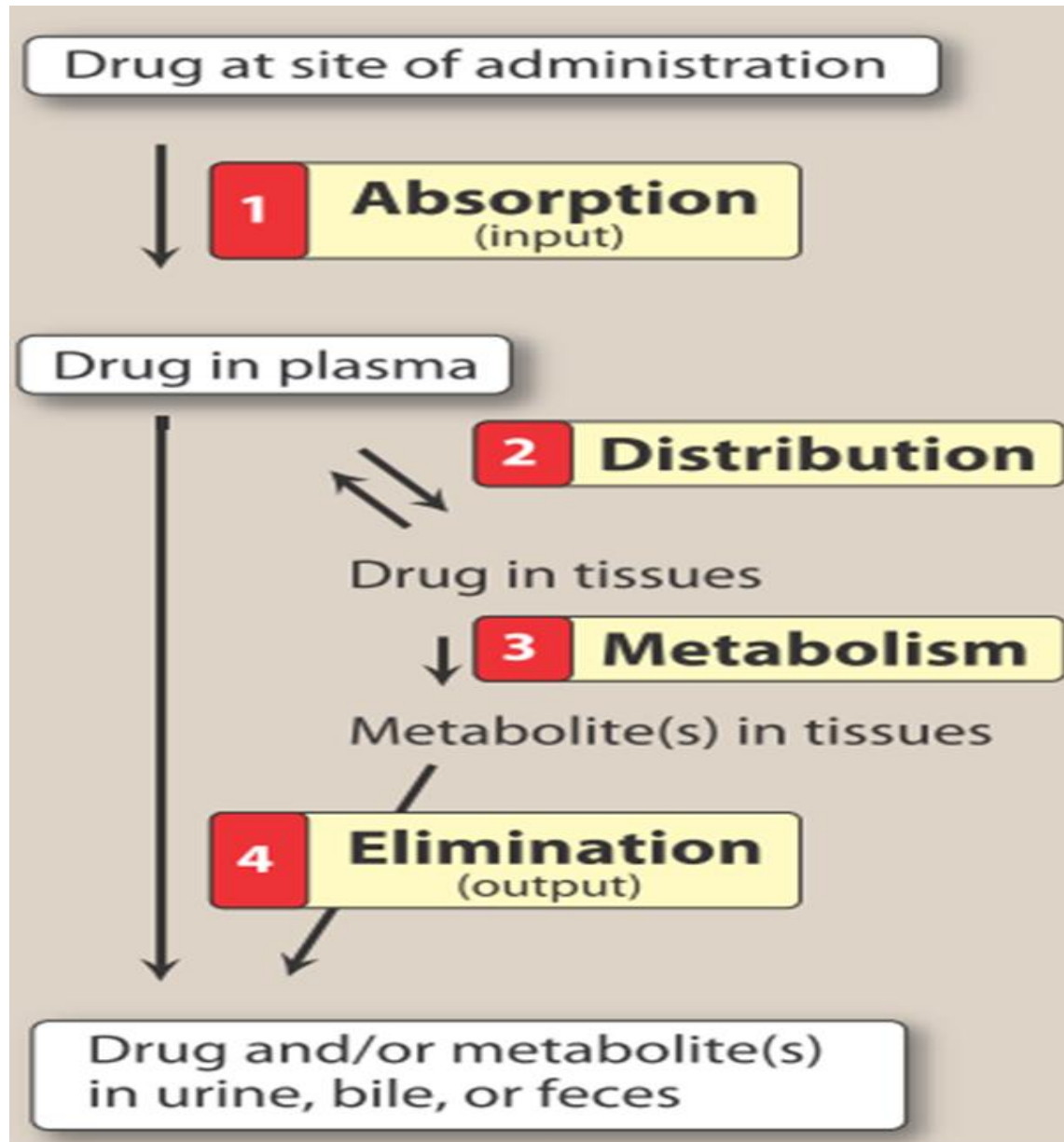
- Dosing intervals
- Duration of action
- Time to reach steady-state
- Risk of accumulation or toxicity

Drugs in Dentistry: Half-life and Clinical Implications



Drug Name	Approximate Half-life	Clinical Implication
Lidocaine	1.5–2 hours	Rapid onset and short duration; ideal for local anesthesia
Amoxicillin	1–1.5 hours	Requires multiple daily doses; effective for dental infections
Metronidazole	~8 hours	Suitable for anaerobic infections; twice-daily dosing
Ibuprofen	2–4 hours	Frequent dosing needed for post-operative pain
Paracetamol	2–3 hours	Safe analgesic; short half-life limits duration

Pharmacokinetics: effects of the body on drugs



ABSORPTION OF DRUGS

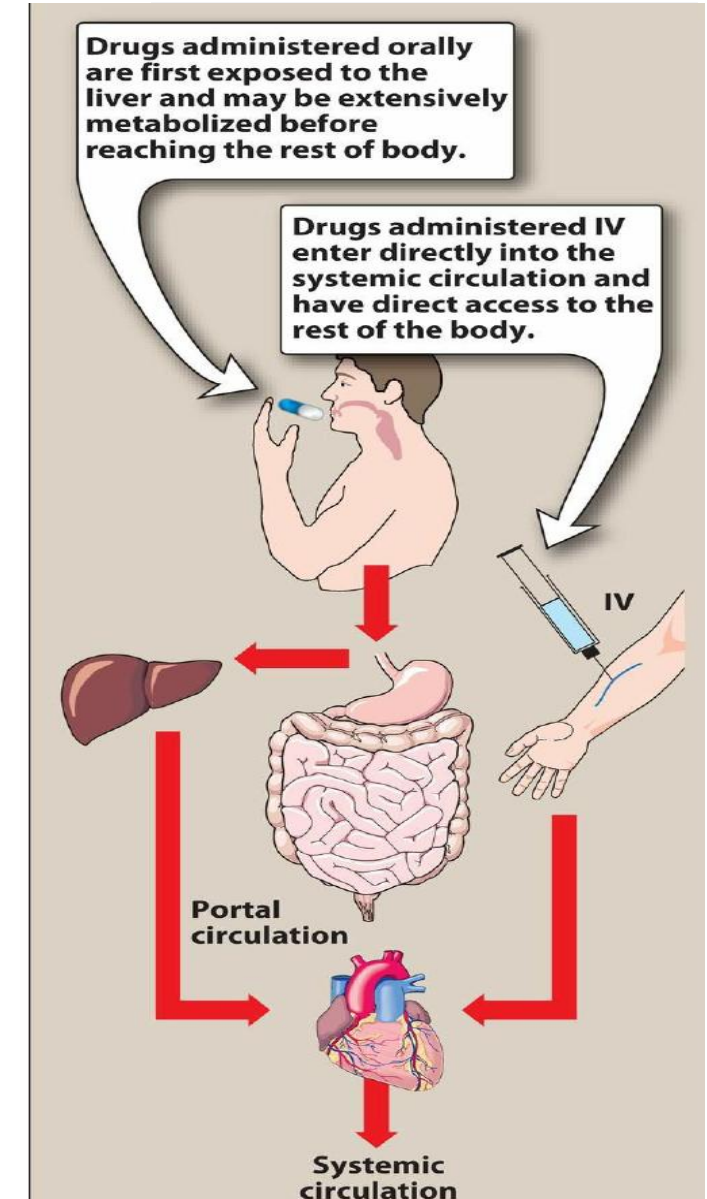
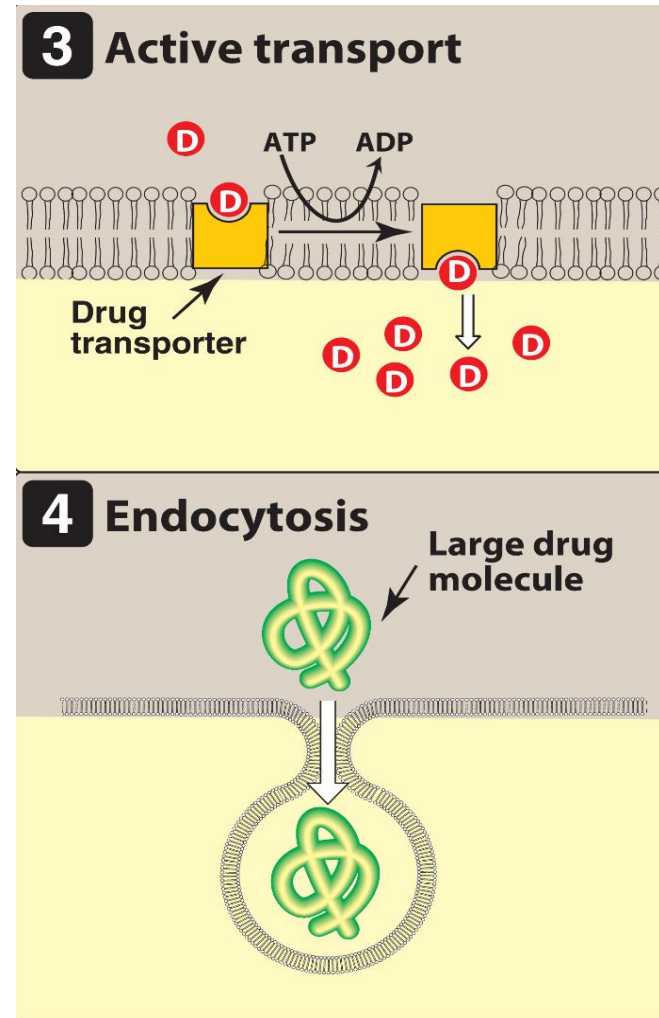
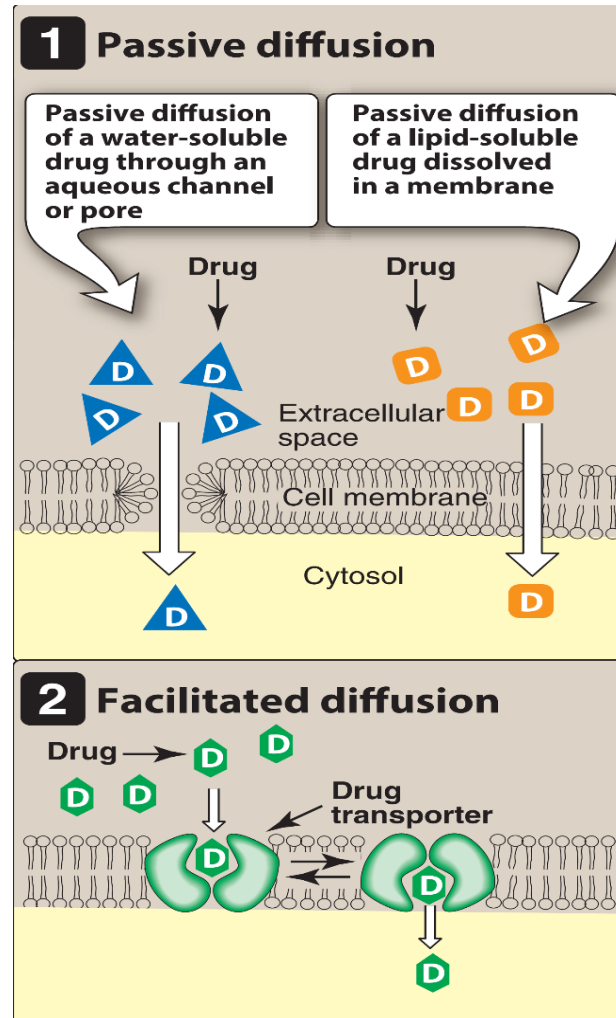
Drug absorption from the gastrointestinal (GI) tract involves several key mechanisms that determine how a drug moves from the site of administration into systemic circulation.

Here's a structured overview:



Mechanism	Description	Characteristics
Passive Diffusion	Drug moves across cell membranes down its concentration gradient.	Most common; depends on lipophilicity, size, and ionization.
Facilitated Diffusion	Carrier-mediated transport down a concentration gradient.	No energy required; selective for certain drug, .e, .glucose
Active Transport	Carrier-mediated transport against a concentration gradient using energy.	Requires ATP; selective and saturable. Vitamins, Amino acid
Endocytosis/Pinocytosis	Drug engulfed by cell membrane, forming vesicles.	Important for large molecules like proteins.

ABSORPTION OF DRUGS



Factors Effecting GI Absorption

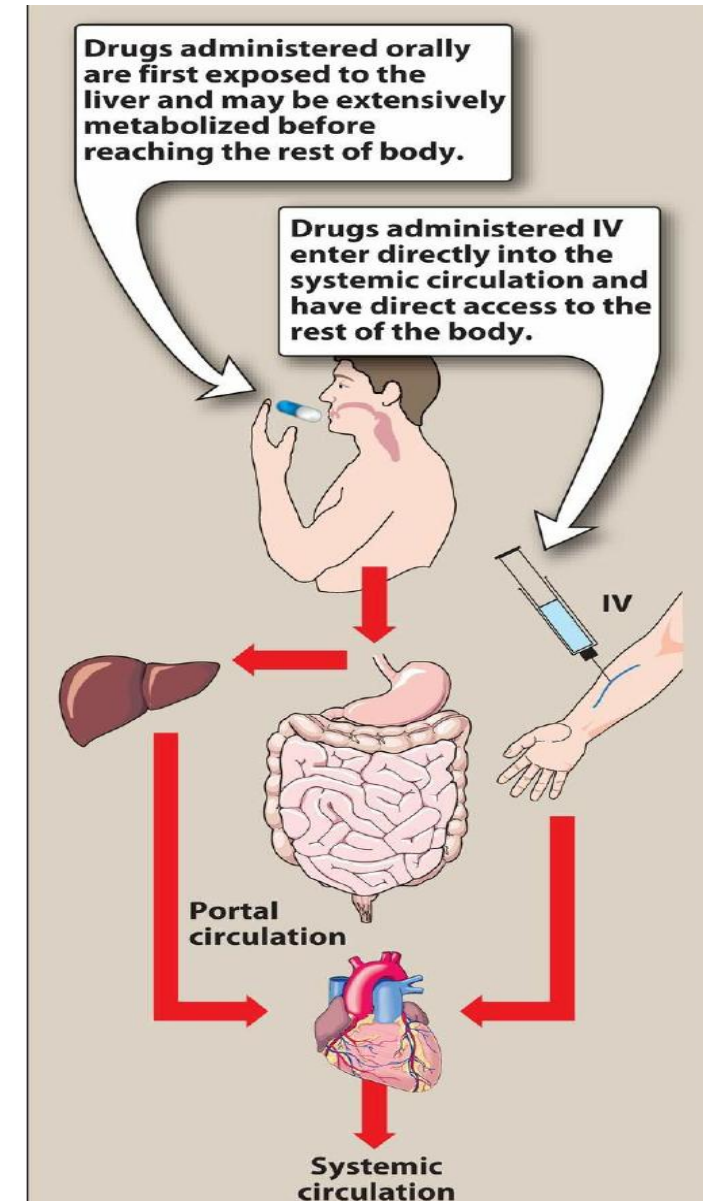


- **Physicochemical properties of the drug:** Solubility, pKa, molecular size, and lipophilicity.
- **Formulation factors:** Dosage form, excipients, and dissolution rate.
- **GI physiology:**
 - Surface area (highest in the small intestine due to villi and microvilli)
 - pH (varies from acidic in the stomach to neutral/alkaline in the intestines)
 - Gastric emptying and intestinal motility
 - Presence of food or enzymes
- **First-pass metabolism:** Drugs absorbed from the GI tract pass through the liver before reaching systemic circulation, which may reduce bioavailability

Bioavailability

Bioavailability refers to the **fraction (%)** of an administered dose of a drug that reaches the **systemic circulation** in its active form.

- For **intravenous (IV) administration**, bioavailability is **100%**.
- For **oral, subcutaneous, intramuscular, or other routes**, bioavailability is usually **less than 100%** due to factors like incomplete absorption or **first-pass metabolism**.



Factors Affecting Bioavailability



Factor	Impact on Drug Absorption
Drug solubility	Poor solubility reduces absorption.
It can	Can enhance or hinder dissolution.
GI pH and enzymes	May reduce or ionize the drug.
First-pass metabolism	Reduces active drug before reaching circulation.
Food interactions	May increase or decrease absorption.
Transport proteins	Efflux pumps (e.g., P-glycoprotein) may reduce absorption.

Drug Distribution

Drug distribution refers to the process by which a drug reversibly leaves the bloodstream and enters the tissues and organs after absorption.

It determines:

- **Where** the drug goes in the body
- **How much** reaches the target site
- **How long** it stays active



Volume of Distribution (Vd)

• A theoretical volume that relates the amount of drug in the body to its concentration in plasma.

- **Low Vd** (e.g., 4–8 L): Drug stays in plasma (e.g., warfarin).
- **High Vd** (>40 L): Drug distributes into tissues (e.g., amiodarone, chloroquine).

$$(Vd) = \frac{\text{Amount of drug in body}}{\text{Plasma drug concentration}}$$

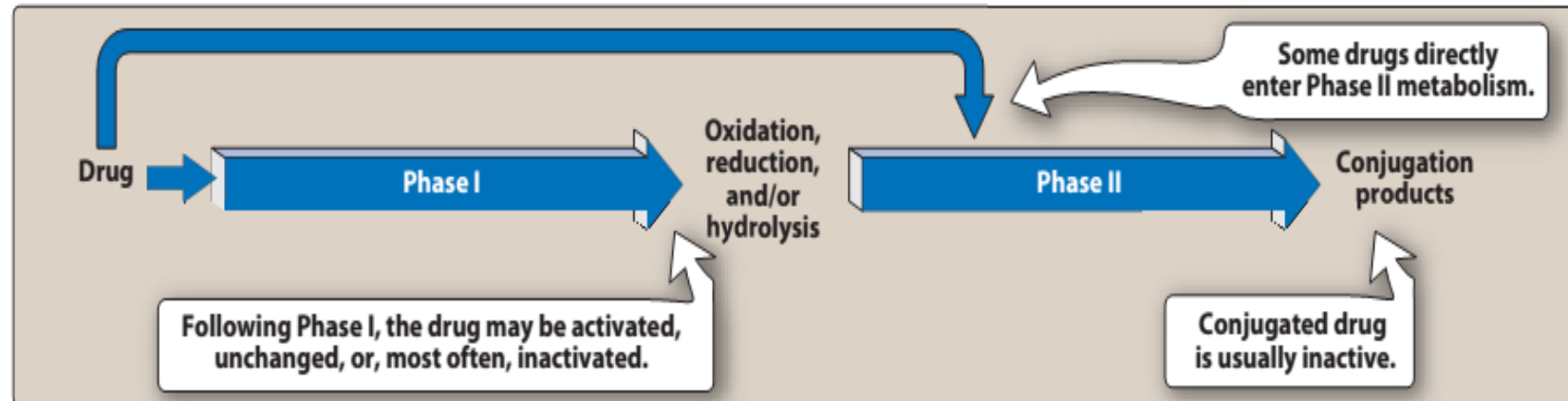
Affecting Effecting Distribution



Factor	Description
Blood flow to tissues	Highly perfused organs (liver, kidney, brain) receive drugs faster.
Plasma protein binding	Drugs bound to albumin stay in plasma; only free drug is active.
Tissue binding	Some drugs accumulate in fat, bone, or muscle (e.g., digoxin in heart).
Lipid solubility	Lipophilic drugs cross membranes easily and distribute widely.
pH and ionization	Ionized drugs are less permeable across membranes.

Drug metabolism

It is the biochemical modification of pharmaceutical substances by living organisms, usually through specialized enzymatic systems. It transforms lipophilic drugs into more hydrophilic forms to facilitate **excretion**.



- **Phase I** often makes drugs more reactive.
- **Phase II** usually detoxifies and prepares for renal/biliary excretion.

Drug Elimination



Drug elimination is the process by which a drug is removed from the body. It determines

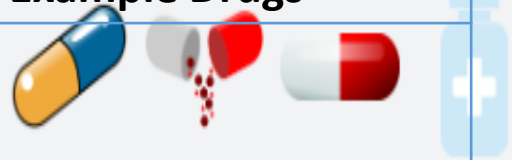
the **duration of drug action** and influences **dosing intervals**.

Clinical Importance

- **Renal or hepatic impairment** slows elimination → risk of accumulation and toxicity.
- **Drug interactions** may alter elimination (e.g., competition for transporters).
- **Age-related changes** affect clearance (especially in neonates and elderly).
- **Therapeutic drug monitoring** is essential for drugs with narrow therapeutic windows.



Route of administration

Route	Description	Advantages	Limitations	Example Drugs
Oral	Swallowed and absorbed via GI tract	Convenient, Low risk of infections (unlike injecting drugs). Toxicities and overdose by the oral route may be overcome	First-pass metabolism, slower onset	 Paracetamol, Ibuprofen
Sublingual/Buccal	Under tongue	Rapid absorption, bypasses liver	Limited to small, lipophilic drugs	Nitroglycerin
Rectal	Inserted into rectum	Useful in vomiting/unconscious patients	Variable absorption, patient discomfort	Diazepam, Paracetamol
Intravenous (IV)	Directly into bloodstream	Immediate effect, 100% bioavailability	Aggressive, risk of infection	Morphine, Antibiotics
Intramuscular (IM)	Injected into muscle tissue	Faster than oral, depot formulations	Painful, variable absorption	Penicillin G, Vaccines
Subcutaneous (SC)	Injected under the skin	Sustained release, self-administration	Limited volume, slower onset	Insulin, Heparin

Other Specialized Routes



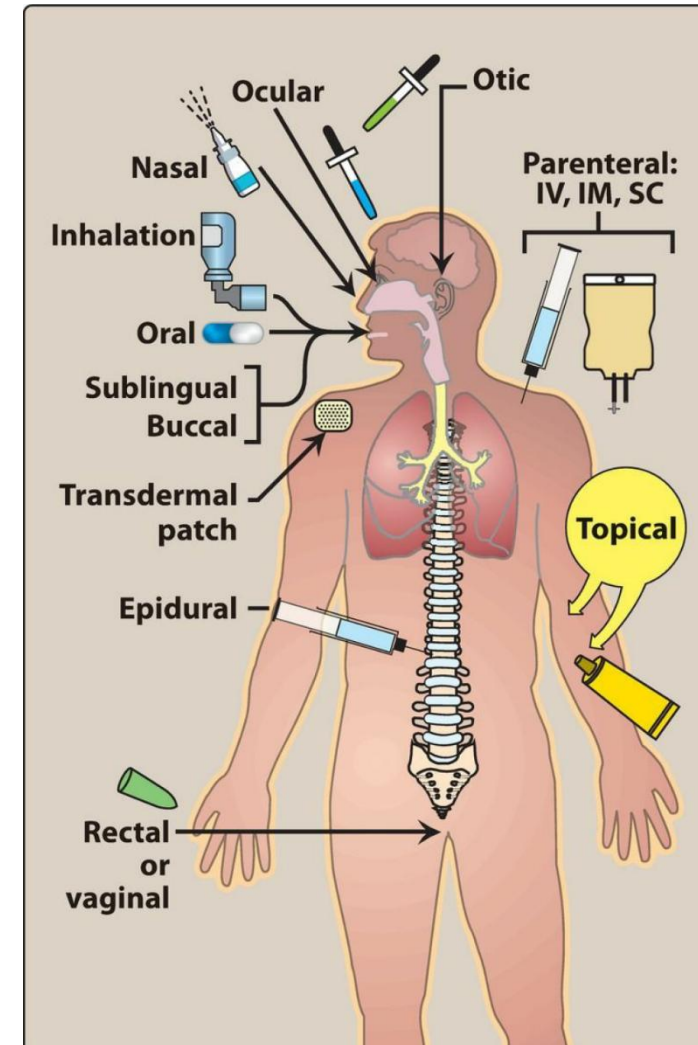
Route	Description	Advantages	Limitations	Example Drugs
Inhalation	Drug inhaled into lungs	Rapid onset, local/systemic effect	Technique-dependent, irritation possible	Salbutamol, Anesthetics
Topical/Transdermal	Applied to skin or mucosa	Local effect, avoids systemic side effects	Limited penetration, slow onset	Lidocaine, Nicotine patch
Intrathecal/Epidural	Into spinal canal or epidural space	Direct CNS effect, potent analgesia	Requires expertise, risk of complications	Bupivacaine, Fentanyl
Ophthalmic /Otic/Nasal	Applied to eyes, ears, or nose	Localized action, minimal systemic effect	Short duration, frequent dosing	Eye drops, Nasal sprays

I- Route of administration (SYSTEMIC EFFECT):

A-Enteral route (administering drug to circulation by mouth):

It is the safest and most common, convenient, and economical method.

- 1-Oral route (swallowed drug)
2. Sublingual (buccal therapy)



I- Route of administration (SYSTEMIC EFFECT):

A-Oral route (swallowed drug):

Advantages:

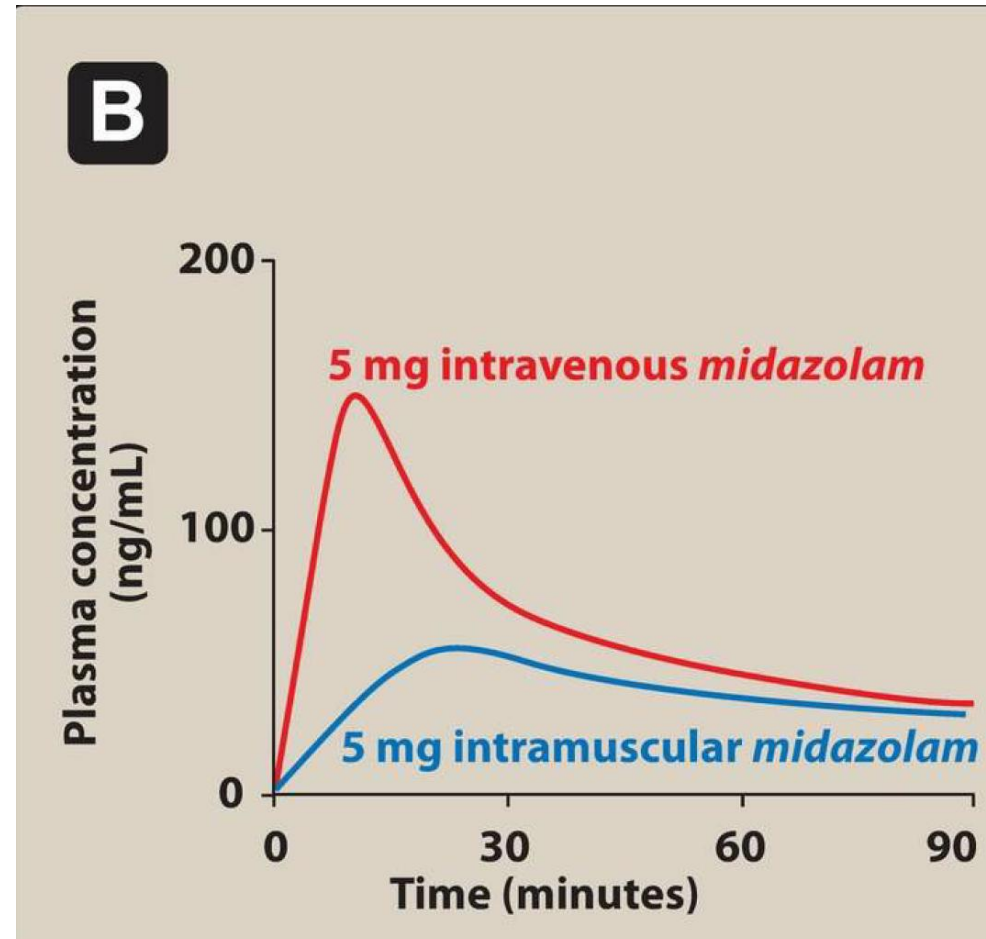
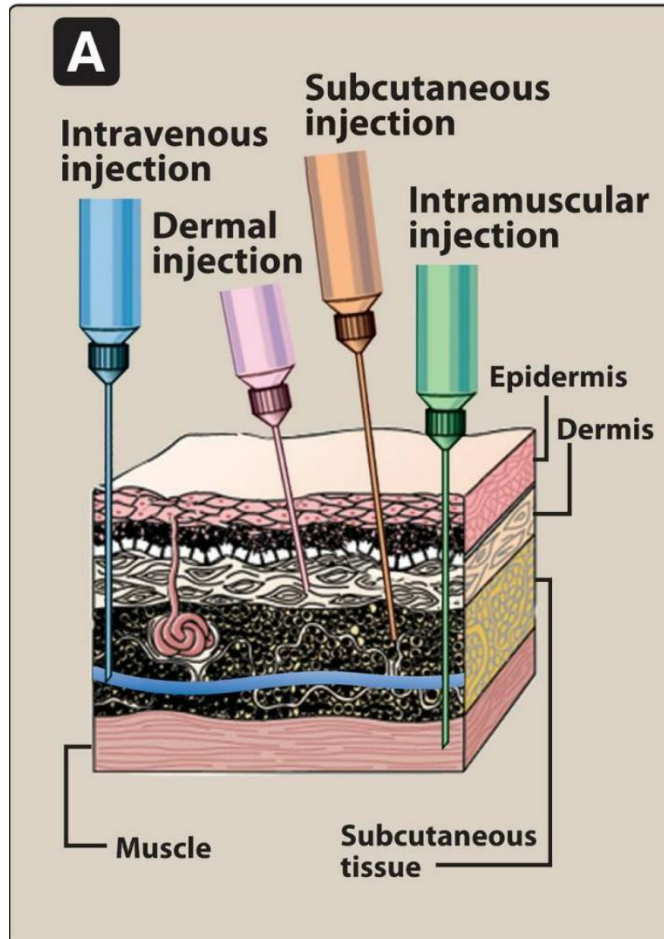
- Easily self-administered.
- Low risk of infections (unlike injecting drugs).
- Toxicities and overdose by the oral route may be overcome with adsorbent antidotes like activated charcoal. Or by GIT decontamination (e.g. Emesis or Gastric lavage)

Disadvantages:

- Drug absorption is more complicated.
- Low pH of the stomach may inactivate some drugs.

Dosage form preparations used could be liquids (e.g. syrup, suspension, elixir ..etc.) or solid forms (e.g. tablet, capsule, ..etc.)

B- Parenteral Administration:



B- Parenteral Administration:

Indications: It is used for drugs that are poorly absorbed from the GI tract (e.g., heparin) or unstable in the GI tract (e.g., insulin) and for patients who are unconscious or under circumstances that require a rapid onset of action.

Advantages:

have the highest bioavailability and drugs are not subject to first-pass metabolism or harsh GI environment.

Disadvantages:

pain, fear, local tissue damage, & infections.

C- other routes:

1.Oral inhalation and nasal preparations

- Provide rapid drug delivery across the large surface area of the mucous membranes of the respiratory tract and pulmonary epithelium
- This route is effective and convenient for patients with respiratory diseases {such as asthma or chronic obstructive pulmonary disease (COPD)}

2.Intrathecal/intraventricular

introduce drugs directly to the cerebrospinal fluid. E.g., intrathecal amphotericin B is used in treating cryptococcal meningitis (because BBB limits drug absorption)

Route of administration contd.

C- other routes:

3.Topical:

- It is used when a local effect of the drug is desired. For example, applying cream directly to the skin, the use of nasal drops, eye drops, scalp lotion or shampoo, antiseptics, powder....etc.



4.Transdermal:

- This route achieves systemic effects by the application of drugs to the skin, usually via a transdermal patch.
- This route is used for sustained delivery of drugs, such as the antianginal drug nitroglycerin and nicotine transdermal patches, which are used to facilitate smoking cessation.



C- other routes:

Rectal:

- ❖ The rectal route is also useful if the drug induces vomiting when given orally or if the patient is already vomiting (e.g. paracetamol suppositories).



Pharmacodynamics

- **Effects of drugs on the body:**
 - **Therapeutic effect** (effect of the drug on the therapeutic target)
 - **Side effect or adverse effect** (effect of the drug on other targets). Sometimes the side effects have clinical benefits, like the sedative effect of some antihistamines.
 - **Toxic effect** (harmful effects, always unwanted and should be avoided during therapy)
 - **Lethal effect** (death)
- **The science that studies the harmful effects of drugs known as Toxicology**

Pharmacodynamics

- Drug effects are produced through chemical actions.
- **How drugs acts?**
- Drug molecules must exert some chemical influence on one or more cell constituents in order to produce a pharmacological response.
- Drug molecules must be **bound to specific targets** in the cells to produce the pharmacological effect.
- ((exceptions include; antacids and osmotic diuretics))

Targets for drug action



• Drugs typically exert their effects by binding to **specific proteins**, leading to a **conformational change** that alters cellular function.

• **Receptor families** include:

- **Ligand-gated ion channels** (e.g., nicotinic ACh receptors)
- **G-protein coupled receptors (GPCRs)** (e.g., muscarinic receptors)
- **Enzyme-linked receptors** (e.g., insulin receptor)
- **Nuclear receptors** (e.g., steroid hormone receptors)

These categories are essential for understanding drug mechanisms, therapeutic effects, and side effects

Primary Targets for Drug Action

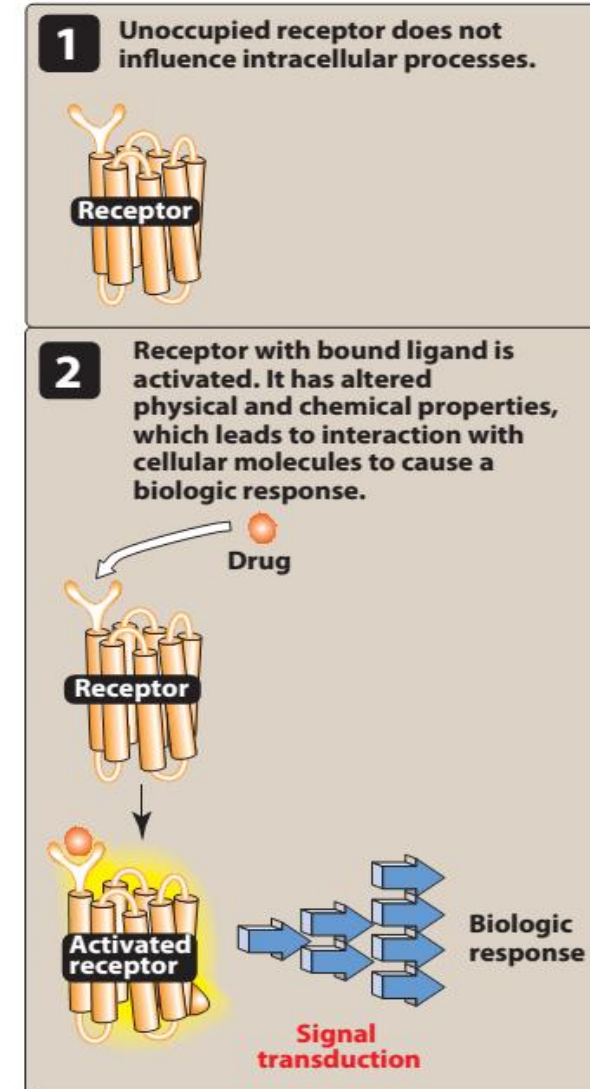


Target Type	Description	Examples
Receptors	Proteins that bind signaling molecules (ligands) to initiate cellular responses	β -adrenergic receptors, opioid receptors
Ion Channels	Gateways for ions across membranes, modulated by drugs to alter cell excitability	Calcium channels, sodium channels
Enzymes	Catalysts for biochemical reactions; drugs may inhibit or activate them	ACE inhibitors, statins
Transporters (Carriers)	Proteins that move substances across membranes; drugs may block or mimic them	SSRIs (serotonin reuptake inhibitors), diuretics

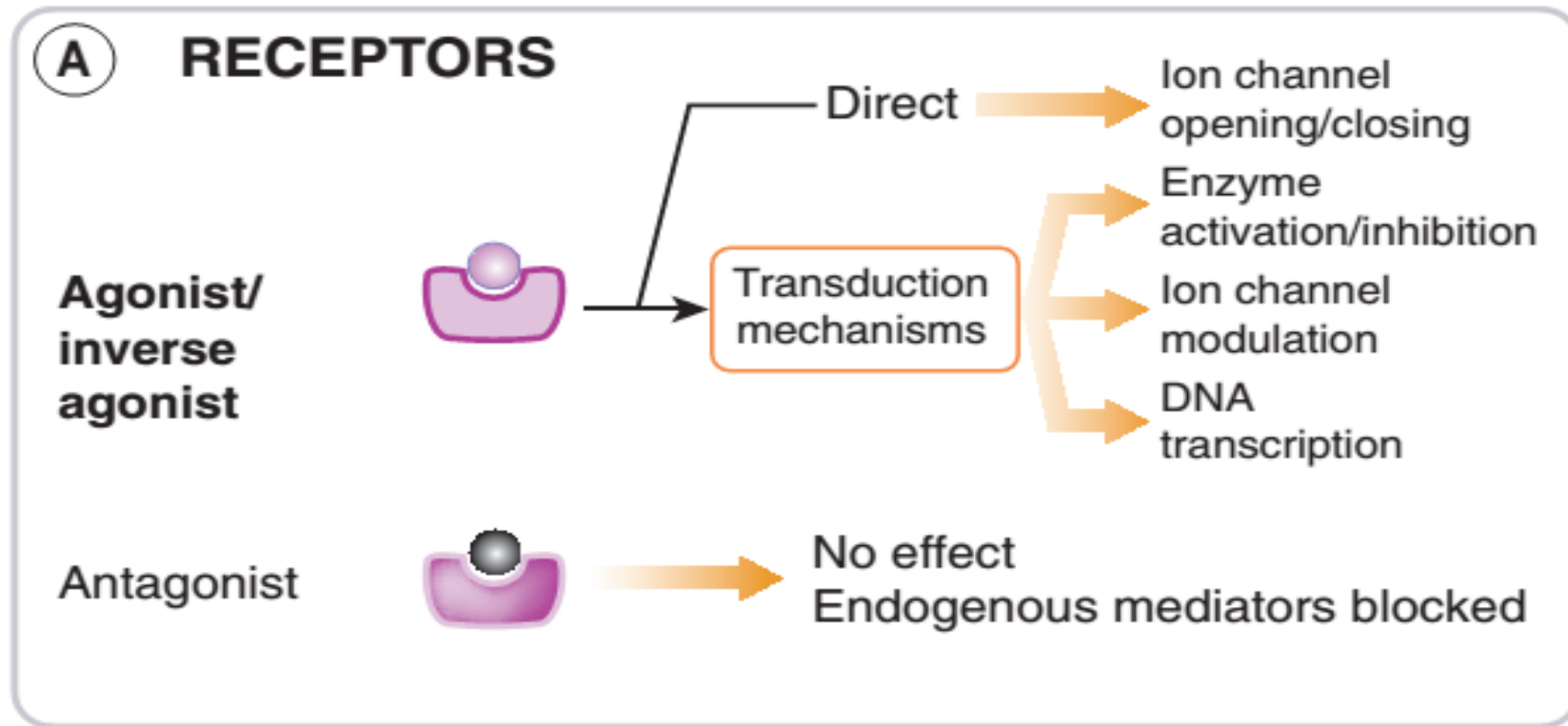
Receptors:

Receptors are specialized target macromolecules (usually proteins) present on the cell surface or within the cell to which a drug binds to initiate its effects.

The term “**ligand**” refers to a small molecule that binds to specific site on the receptor and activates the receptor (the ligand may be a naturally occurring molecule or a drug).



Receptors



E.g. β - receptors:

Agonist = adrenaline

Antagonist = propranolol (β -blocker)

Inverse agonist = β - Carbolines

Agonist:

An agonist drug can bind to a receptor and produces a biological response.

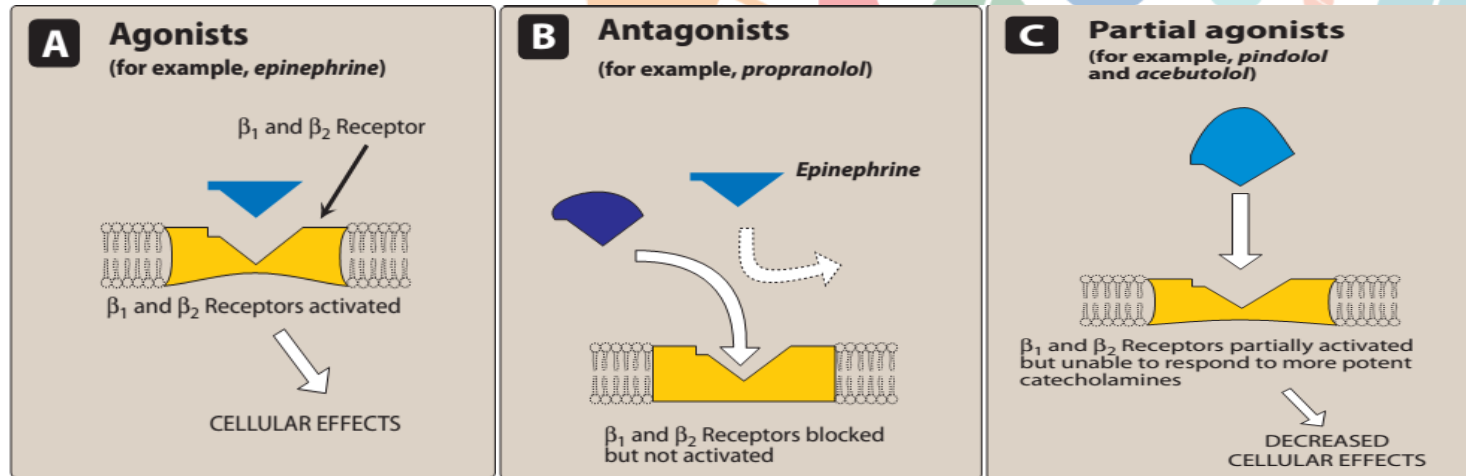
It's the drug that resembles the natural transmitter or hormone, may activate the concerned receptor and result in a response, so an agonist has "affinity" as well as "intrinsic activity"

((The capacity of a drug to interact with a receptor)) = affinity.

((Capability to produce a response)) = intrinsic activity

e.g.: Noradrenaline, Acetylcholine, and their chemical analogs

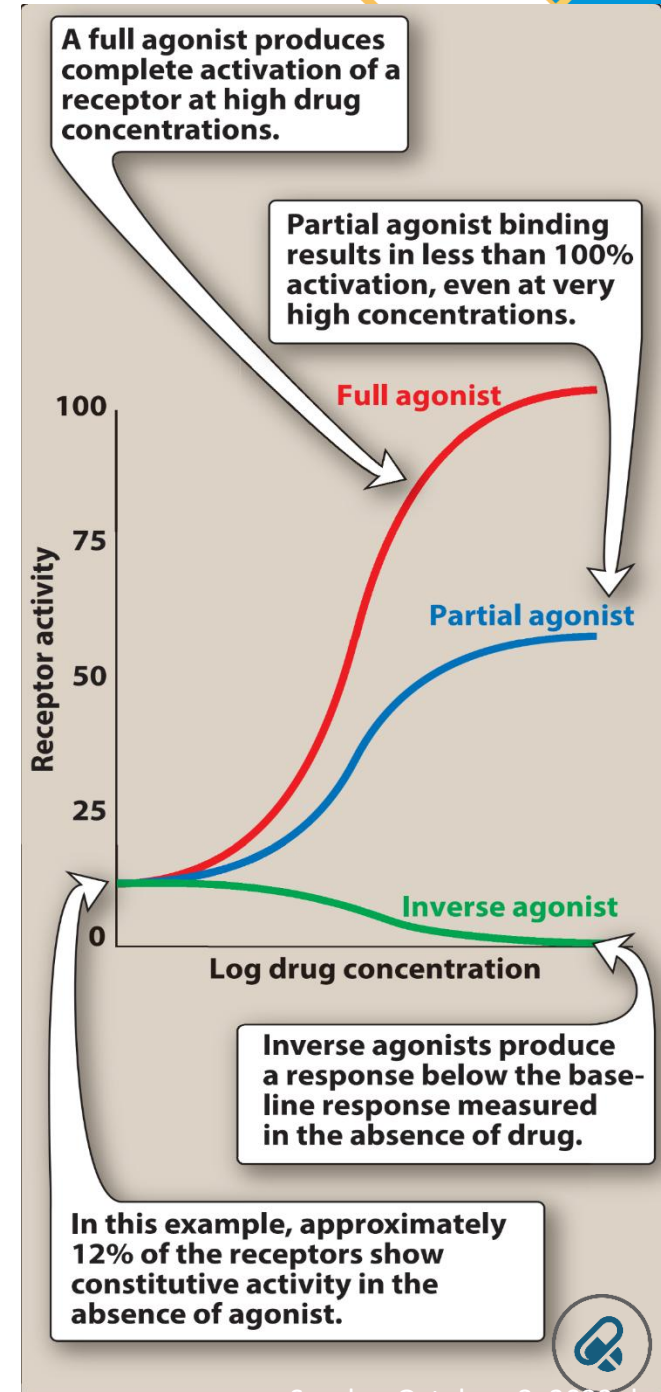
Agonist:



Agonist (Types of agonist)

Full agonists:

- The drug binds to a receptor and produces a maximal biologic response (100% efficacy or E_{max}) that mimics the response to the endogenous ligand like; adrenaline is a full agonist on β -adrenoceptors.
- In general, a full agonist has a strong affinity for its receptor and maximum efficacy.

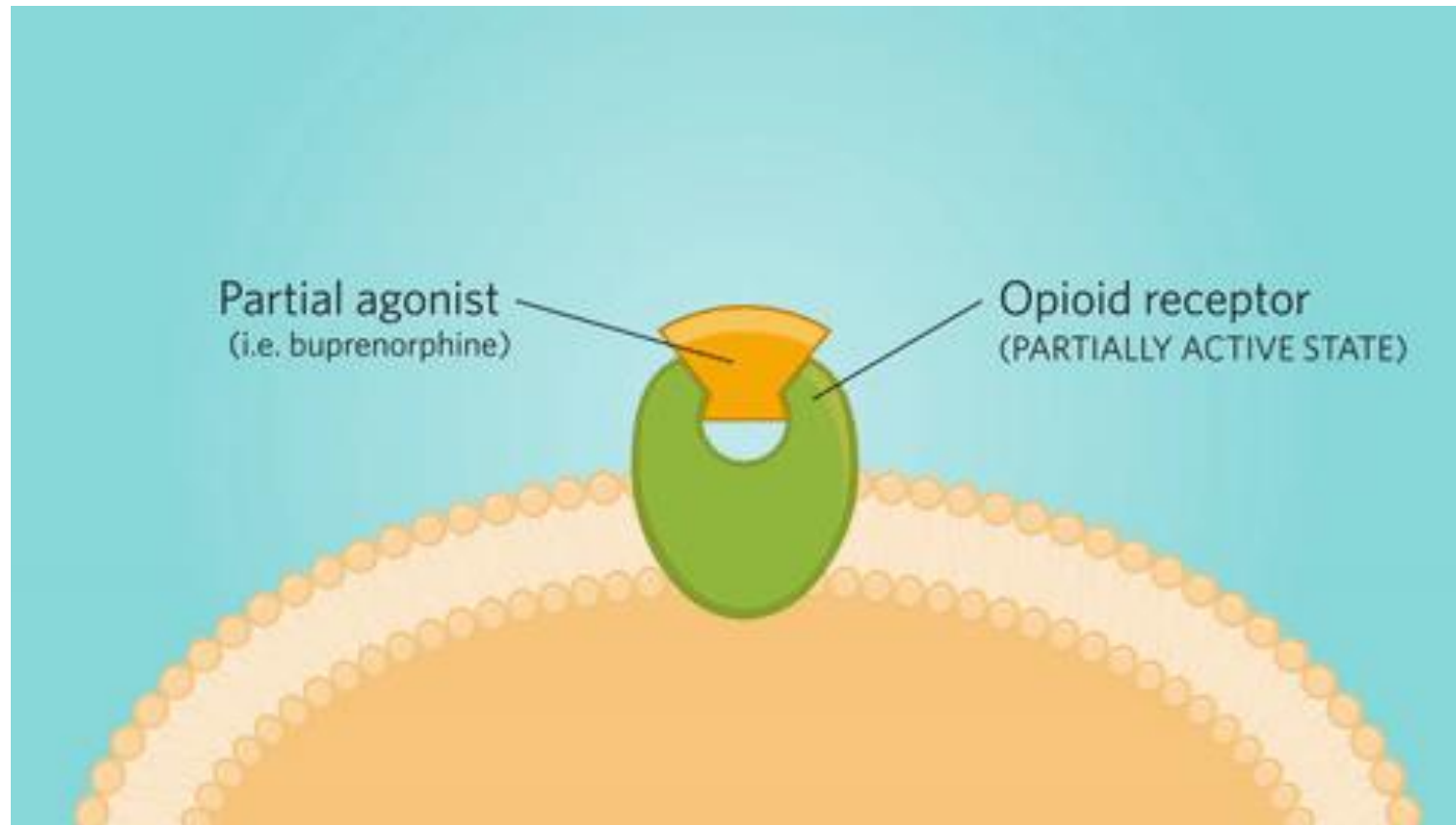


Agonist (Types of agonist)

Partial agonists:

- Have efficacy (intrinsic activity) greater than zero but less than that 100%.
- Partial agonist may have an affinity that is greater than to that of a full agonist.
- A unique feature of these drugs is that, under appropriate conditions, a partial agonist may act as an antagonist of a full agonist e.g. pindolol has partial agonist effect on β -adrenoceptors and acts as antagonist in the presence of excess adrenaline (full agonist) during exercise, but can stimulate the β -receptors at rest (partial activation).

Partial agonist



Agonist (Types of agonist)

Inverse agonists:

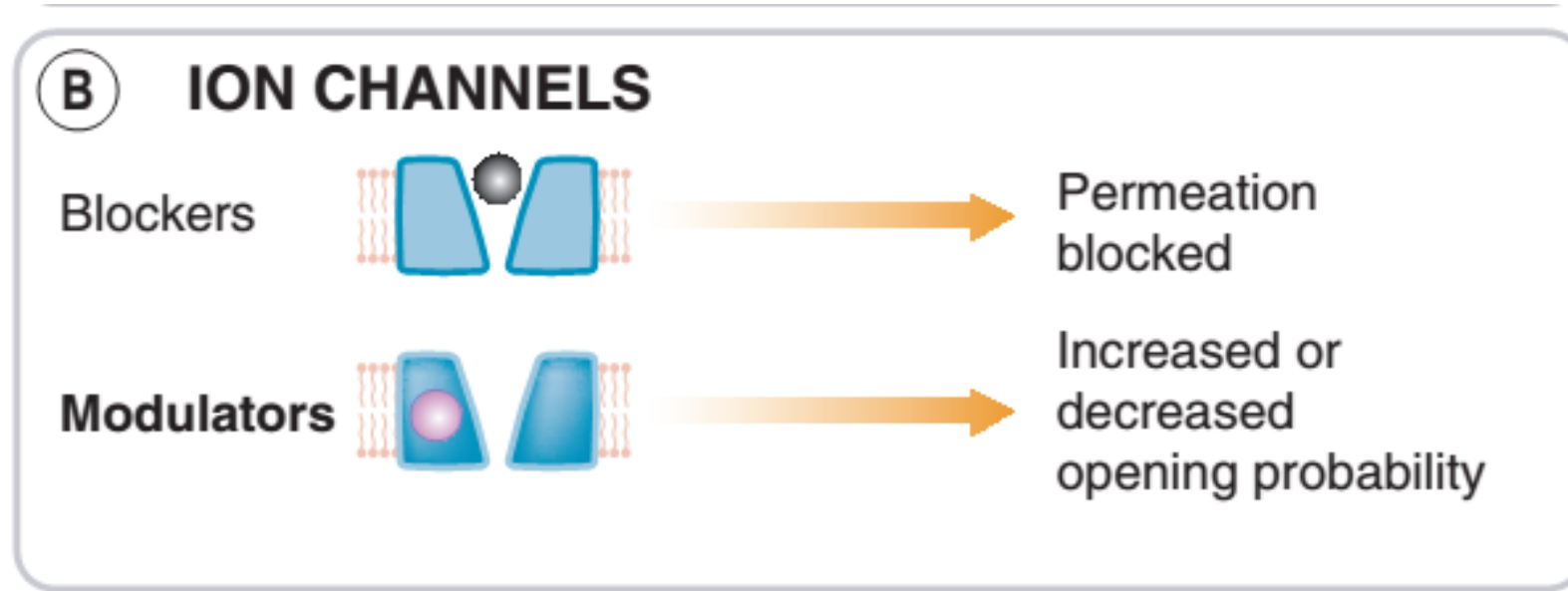
- A drug that binds to the same receptor of agonist but induces a pharmacological response opposite to that agonist. It stabilizes the inactive R form of the receptor.
- An inverse agonist has efficacy less than the baseline efficacy measured in the absence of an agonist. E.g., β -Carbolines on benzodiazepine receptors.



ANTAGONIST Drugs:

- **Antagonists** are drugs that bind to receptors and produce no effect if an agonist is not present (block the receptors). They can decrease or oppose the actions of **another drug or endogenous ligand (agonist)**.
- Antagonists have no intrinsic activity (zero or no efficacy) and, therefore, produce no effect by themselves, but they have **strong affinity**.
- **Propranolol** is antagonist on β -adrenoceptors.

Ion channels

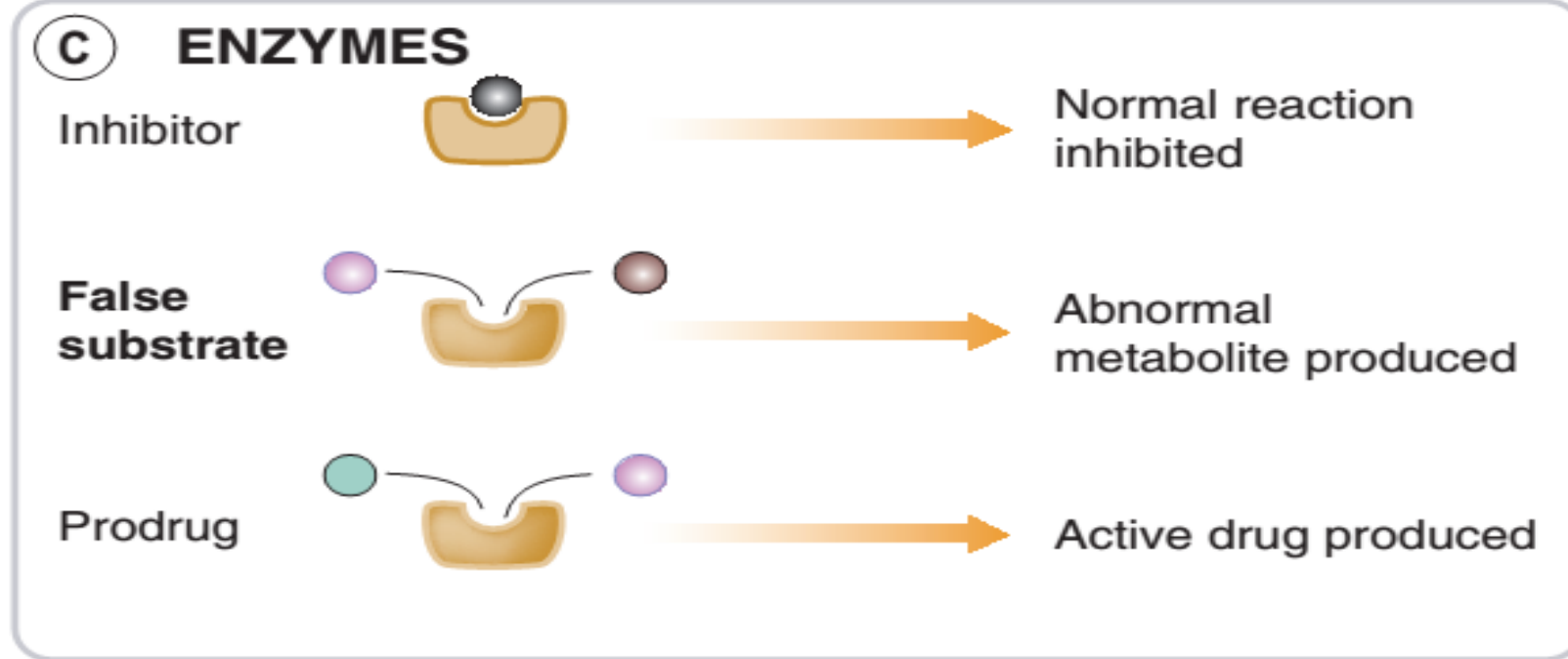


E.g. voltage-gated Na⁺ channels:

Blocker = lidocaine (local anesthesia)

Modulator = phenytoin (antiepileptic)

Enzymes

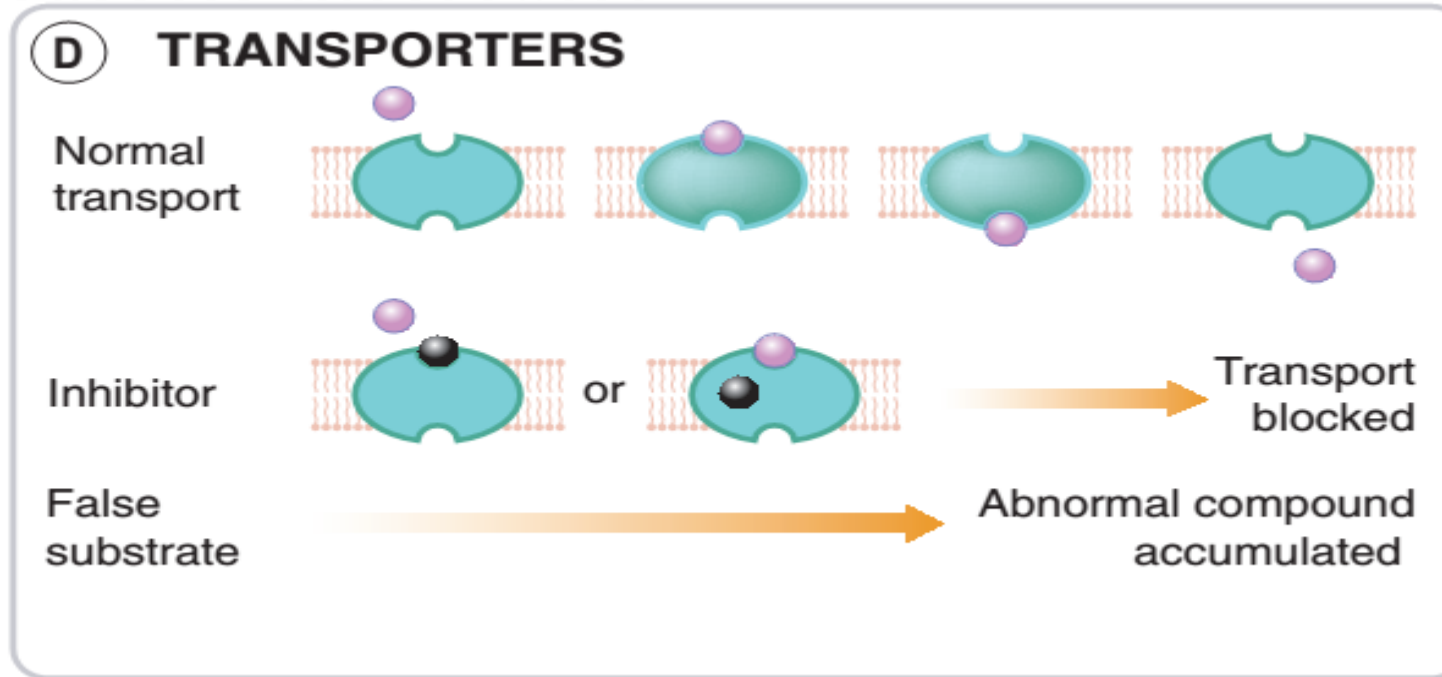


Inhibitor (aspirin inhibit cyclooxygenase enzymes → decrease prostaglandins & thromboxane production)

False substrate (methyldopa metabolized by DOPA decarboxylase enzyme into methyldopamine “false” transmitter)

Prodrug (Azathioprine is inactive immunosuppressant drug, but converted by liver enzymes into an active drug; 6-mercaptopurine)

Carrier molecules



Inhibitor (Cocaine inhibit noradrenaline transporter → prevention of reuptake → postsynaptic accumulation → CNS stimulation)

False substrate (Amphetamine transported (by reuptake) instead of noradrenaline and accumulated presynaptically → postsynaptic accumulation of noradrenaline → CNS stimulation)

Drug - Receptor Interaction

- The binding of drugs to receptors, cause receptors to change from an inactive state (R) to an activated state (R*). The magnitude of the response is proportional to the number of drug-receptor complexes:

Drug + Receptor $\rightleftharpoons$ Drug-receptor complex $\rightarrow$ Biologic effect



Affinity: the tendency of a drug to bind to receptor, the higher the affinity, the lower the concentration required to produce a given level of receptor occupancy. Several types of bonds are involved in drug - receptor interaction:

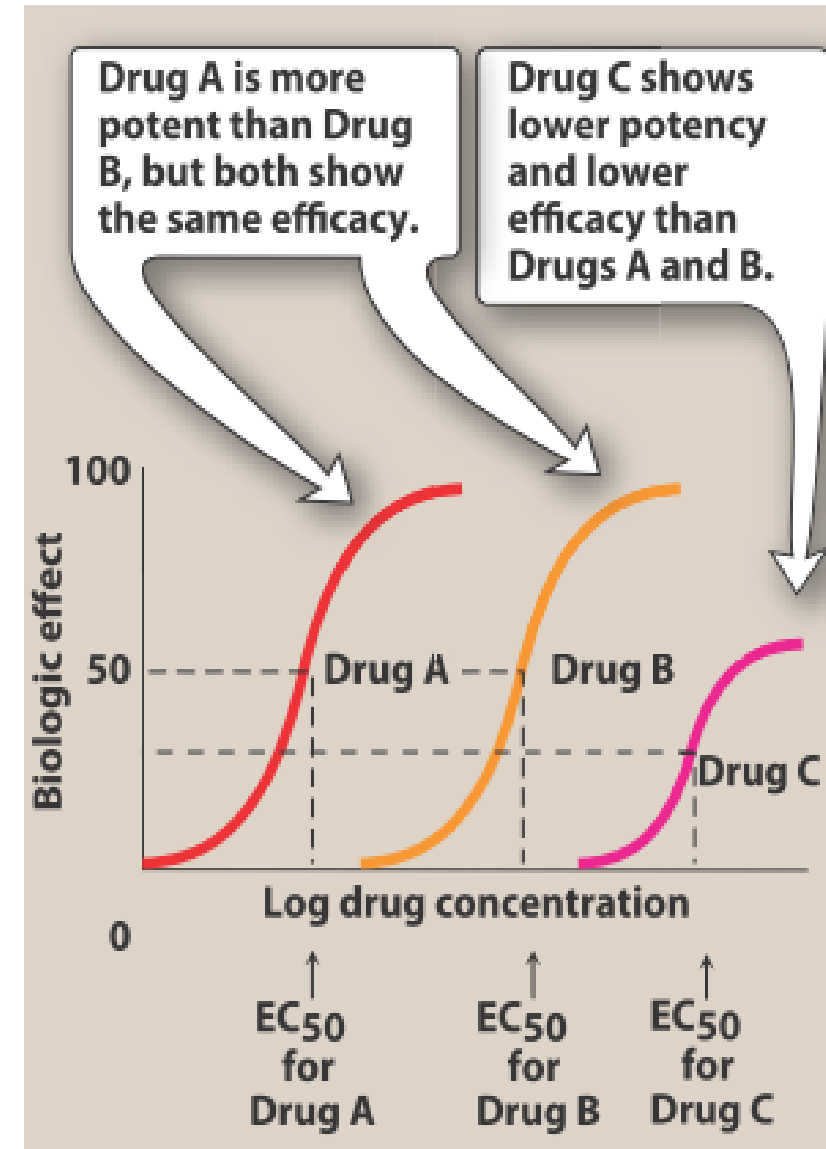
- Van der Waals bonds
 - Hydrogen bonds
 - Electrostatic bonds
 - Covalent bonds
- Irreversible binding
- Reversible binding

Potency: the amount of drug necessary to produce an effect of a given strength. The concentration of drug-producing 50% of the maximum effect is used to determine potency and is commonly designated as the EC₅₀.

Efficacy: is the ability of a drug to elicit a response when it interacts with a receptor (efficacy = strength of response).

The maximal efficacy is referred to (E_{max}).

Efficacy is more important than drug potency. A drug with greater efficacy is more therapeutically beneficial than one that is more potent.



Pharmacology

- Dental Pharmacology (basic & clinical pharmacology for dentists)
- *During this course you will study;*
- 1- General pharmacology:
 - ❖ Pharmacokinetics
 - ❖ Pharmacodynamics
- 2- Systemic pharmacology:
 - ❖ Drugs affecting body systems closely linked to dentistry like, ANS, CNS, CVS,etc.

Pharmacology

- For each drug (or group of drugs) studied, you should know the followings:
 - Classification (drug class)
 - **Mechanism of action**
 - Pharmacokinetics (administration, $t_{1/2}$, metabolism,etc.)
 - **Pharmacological effects**
 - Clinical use (indications)
 - **Adverse effects and toxicity**
 - Contraindications and precautions
 - Drug interactions
 - **Any specific information related to drug use.**



References



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- **Katzung's** Basic & Clinical Pharmacology, 14th Edition. 2018
- **Rang and Dale's** Pharmacology, 8th Edition. 2021



THANK
YOU!

